

- Compliant with NIST FIPS publication 197 “Advanced encryption standard (AES)” (November 2001)
- Encryption and decryption with multiple chaining modes:
 - Electronic codebook (ECB) mode
 - Cipher block chaining (CBC) mode
 - Counter (CTR) mode
 - Galois counter mode (GCM)
 - Galois message authentication code (GMAC) mode
 - Counter with CBC-MAC (CCM) mode
- 128-bit data block processing, supporting cipher key lengths of 128-bit and 256-bit
 - 51 or 75 clock cycle latency in ECB mode for processing one 128-bit block with, respectively, 128-bit or 256-bit key
- Integrated key scheduler to compute the last round key for ECB/CBC decryption
- 256-bit write-only registers for storing cryptographic keys (eight 32-bit registers)
- 128-bit registers for storing initialization vectors (four 32-bit registers)
- 32-bit buffer for data input and output
- Automatic data flow control supporting two direct memory access (DMA) channels, one for incoming data, one for processed data. Only single transfers are supported.
- Data-swapping logic to support 1-, 8-, 16-, or 32-bit data
- AMBA AHB slave peripheral, accessible through 32-bit word single accesses only. Other access types generate an AHB error, and other than 32-bit writes may corrupt the register content.
- Software (in CPU mode only, not in DMA mode) can suspend a message if AES needs to process another message with a higher priority, then resume the original message.

Table 5. AES features

Modes or features ⁽¹⁾	AES
ECB, CBC chaining	X
CTR, CCM, GCM chaining	X
AES 128-bit ECB encryption in cycles	51
DHUK and BHK key selection	-
Resistant to side-channel attacks	-
Shared key between SAES and AES	X
Key sizes in bits	128/256

1. X = supported.

3.21 HASH processor (HASH)

The hash processor is a fully compliant implementation of the secure hash algorithm (SHA-1, SHA-2 family) and the HMAC (keyed-hash message authentication code) algorithm. HMAC is suitable for applications requiring message authentication.

The hash processor computes FIPS (Federal Information Processing Standards) approved digests of lengths of 160, 224, and 256 bits for messages of any length less than 2×64 bits (for SHA-1, SHA-224, and SHA-256).

The HASH main features are: